# Hybrid Deep Learning Architecture With K-means Clustering For Weapon Detection In CCTV Surveillance

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2024 IEEE International Conference on Computer Vision and Machine Intelligence (CVMI) | 979-8-3503-7687-6/24/$31.00 ©2024 IEEE | DOI: 10.1109/CVMI61877.2024.10781588

***Abstract*—For quick criminal activity alert, it became quite obvious to use multiple capture points with cameras, and with multiple capture points, there is need for automated criminal activity alerting systems so the human observer can manage all the CCTV feed in real-time, as its humanly not possible go through that many video feed without a crime slipping out undetected. A deep learning based approach for this task on various network have been researched previously. The researchers have tuned with larger and largest network possible to perform weapon detection for surveillance, though they have achieved more than 90% of accuracy for the task but have to pull largest and complex networks possible. Large deep learning models are costly in both computation and memory for a CCTV device to perform AI workload.There was a gap in studying hybrid approach where deep learning along with machine learning based approach are evaluated for the task. To close this gap, our study employs a hybrid approach that combines machine learning and deep learning methods.Training on a customised dataset was attempted initially. But when implementation proved challenging, the study transitioned to implementing the use of the "OD-weapon detection dataset" that had been collected from GitHub. Different levels of accuracy were achieved on first validation by using deep learning models such as VGG16, VGG19, InceptionNet, and MobileNet, which were maximised by applying this diversified collection of weapon images. Techniques for clustering, fine-tuning, and PCA dimension reduction were used to improve the classification performance.***

***Index Terms*—CCTV footage, weapon detection, VGG16,VGG19,MobiNet,Effeciency**

## I. Introduction

There has been video monitoring in place for over 10 years. Providing the authority responsible for security a live video stream, a viewer could face two challenges.In the beginning, humans' poor capacity to analyse information would cause them to miss possible illegal acts when presented with an excessive number of capture points.For this reason, we need an AI-enabled camera that will notify the observer of any illegal action that involves the use of a detecting weapons. [1].The second need is that edge devices be used for scene comprehension.Artificial intelligence on edge devices. A camera that can detect criminal activity while on the go and stream an alert to an observer is a far better solution than a camera that only relays video feed to an algorithm that runs on the cloud. [2]. This is because the camera can work offline, continuing to do its task even in the event of a power outage or network disruption for a short while. Identifying weapons is a subset of criteria within the broader field of identifying criminal conduct. Therefore, it becomes increasingly vital to investigate networks capable of effectively identifying weapons and whether they can be improved by utilising machine learning methods, which are less expensive to compute than deep learning.

Technological solutions that can efficiently identify the criminal goods in CCTV feeds are desperately needed in considering these challenges. Combining image processing methods with machine learning algorithms—especially deep learning models like convolutional neural networks (CNNs [3])—is one of these innovative and potentially effective approaches. Through the use of CNNs' ability to identify patterns in images, these models can be trained on large datasets that contain both acceptable and unsuitable content. This allows them to correctly identify and remove offensive content. By using this strategy, content moderation efforts may be greatly strengthened, which would benefit all users by creating a more comfortable and safe online environment. This paper proposes a hybrid approach that combines the best features of deep learning and conventional machine learning techniques to create a strong and effective classification framework, with a view of improving the accuracy of the system in identifying weapon

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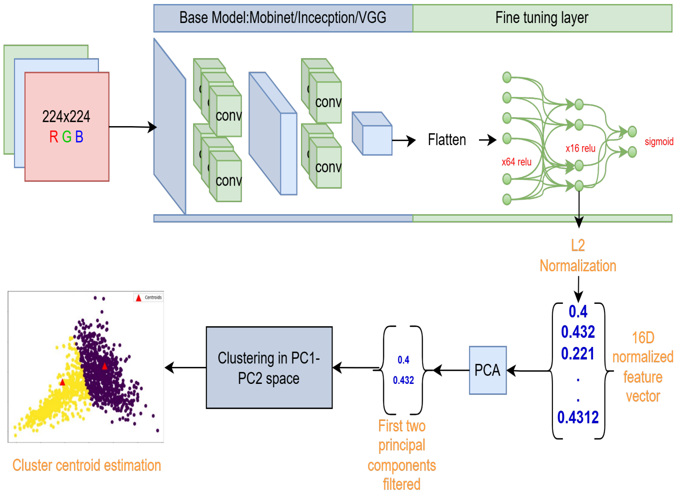


Fig. 1. Overview of approach

imagery and content classification. A complete method for evaluating complex visual data and using its effective feature extraction capabilities is provided by the combination of CNNs with machine learning algorithms. K-means clustering has been employed in order to improve the classification process further. The suggested approach, based on network-extracted characteristics, facilitates more detailed analysis and classification by identifying discrete groups or clusters of related first two principle components. The efficacy of the classification work is increased by employing feature evaluation techniques like homogeneity assessment and dimensional reduction of the retrieved feature using PCA, before clustering. The resulting model as shown in Fig.1, demonstrates proficiency in creating decision boundaries based on the identified clusters,with a specific focus on weapon imagery. This hybrid approach represents a significant advancement in content classification methodologies, offering greater accuracy and efficiency in identifying objectionable material in a CCTV video frame.

## II. Literature

Recent research have demonstrated creative uses of deep learning techniques in the field of improving surveillance and supporting public safety measures. In their research, Dwivedi, Singh, and Kushwaha [4] offer a perceptive method for weapon categorisation, suggesting a Deep Convolutional Neural Network (DCNN)-based system. By using transfer learning and precision-recall analysis, their system—which used the VGGNet architecture—achieved an astounding accuracy level of 98.41Navalgund and Priyadharshini [5] introduce a Crime Intention Detection System utilizing VGGNet-19, which surpasses GoogleNet InceptionV3 in training accuracy[6], offering real-time analysis with SMS alerts for authorities, thus addressing the limitations of conventional CCTV surveillance for crime detection. Thus in a comparative study, Mascarenhas and Agarwal evaluated VGG16[7], VGG19[8] and ResNet50 architectures[9], highlighting ResNet50's superior performances of a 97.33% accuracy rate in real-life image classification tasks, particularly significant in refining pricing comparison mechanisms in retail networks. Additionally, Wong and Lin's investigation into CNN architectures for weapon classification reveals InceptionV3's remarkable accuracy of 99% [10], demonstrating the robustness of CNNs in security and law enforcement contexts, even amidst additional non-weapon classes.

## III. Methodology

In this paper we report study on performance of VGG16,VGG19,MobieNet and IncecptionV3 [6] for weapon detection over OD-Weapon detection dataset [1].The performance evaluation are also done for network utilized with machine learning algorithm.

In order to evaluate each model's performance in identifying weapons in low-resolution video, such as that found in CCTV [2] footage, we methodically tracked the accuracy metric that each model produced. To further throw the light on the advantages and disadvantages of each model, we also use clustering techniques to examine the similarities and contrasts between them. With this method, we may assure the creation of an effective detection system by gaining an adequate understanding of the capacities of various deep-learning architectures in recognising images associated with crime. Furthermore, we added fine-tuning to our analysis.

The used dataset is designed for tasks requiring item detection, with a particular emphasis on the identification of weapons in images. It includes a number of photos with bounding boxes around things that are weapons, such knives and guns, and remarks indicating the weapons' locations inside the images as well as what they are. Each of the 10,080 images in the collection shows people carrying weapons in various situations. 1,511 of them were put apart for testing, while 8,569 of them were used for training. The bounding box coordinates for the weapons that were found and labels indicating the kind of weapon (knife, gun, handheld, etc.) for every image have been included in the dataset's annotations. The annotation for the dataset is written in XML. We kept separate files for annotations for every image.I Table I provides a quick overview of the dataset's training/validation set settings and imagine pre-processing.

TABLE I
PREPROCESSING PARAMETERS

| | |
|---|---|
| Image dimension | 150 * 150 * 3 |
| Training set generator | keras training datagenerator(from folder) |
| Augmentation | yes |
| augmentation parameters | shear ,skew, zoom, height and width shift |
| class mode | categorical |
| batch size | 32,16,25 |
| desired labels | 3 |
| label name | knife ,guns,hand Held weapons |

[1]https://github.com/ari-dasci/OD-WeaponDetection

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The deep learning networks were fine-tuned on the dataset with additional custom layers as shown in Table II whose performance were evaluated using the performance metrics

## IV. Implementation environment and specification

Proposed models were programmed using Tensorflow 2 and Keras.We designed them on Google Colab virtual machine with GPU runtime.The experiment was conducted on following specific hardware environment aquired using neofetch command.

- CPU: Intel Xeon(2) @ 2.199GHz
- RAM: 14GB
- GPU: Nvidia-T4(12GB).
- Storage: Atleast 20GB SSD

The software requirements to be installed are as given below:

- Ubuntu 22.04.3 LTS x86_64
- Python 3.10.12
- Modules ver: Keras 3.4.1, Tensorflow 2.17.0, datasets 2.21.0, numpy 1.26.3

## V. Neural Network constituent

TABLE II
Fine Tuning Layers

| **Base Model** | **MobiNet** | **VGG19** | **VGG16** | **IncecptionV3** |
|---|---|---|---|---|
| weights | imagenet | imagenet | imagenet | imagenet |
| Custom layers | baseModel<br>128,relu<br>16,relu<br>dropout(0.2)<br>3,softmax | baseModel<br>512,relu<br>512,relu<br>dropout(0.3)<br>3,softmax | baseModel<br>512,relu<br>dropout(0.5)<br>3,softmax | baseModel<br>64,relu<br>16,relu<br>dropout(0.2)<br>3,softmax |
| epochs | 20 | 30 | 30 | 30 |

Table II is reference provided for quick capture of our fine tune layers in each model including how much epochs the models were trained while each experiment.This table provides easy and fast capture of model design we experimented.The model's custom layers were empirically selected showing optimal results for fine tuning in respect to their base models.

## VI. Results and discussion

The performance of the deep learning models in detecting weapon was evaluated using the accuracy metric defined in eq.1.

TableIII summarizes the accuracy achieved by each model after fine tuning with custom layers. Prior to integration with clustering and PCA, the deep learning models (VGG16, VGG19, InceptionNet, and MobileNet) underwent

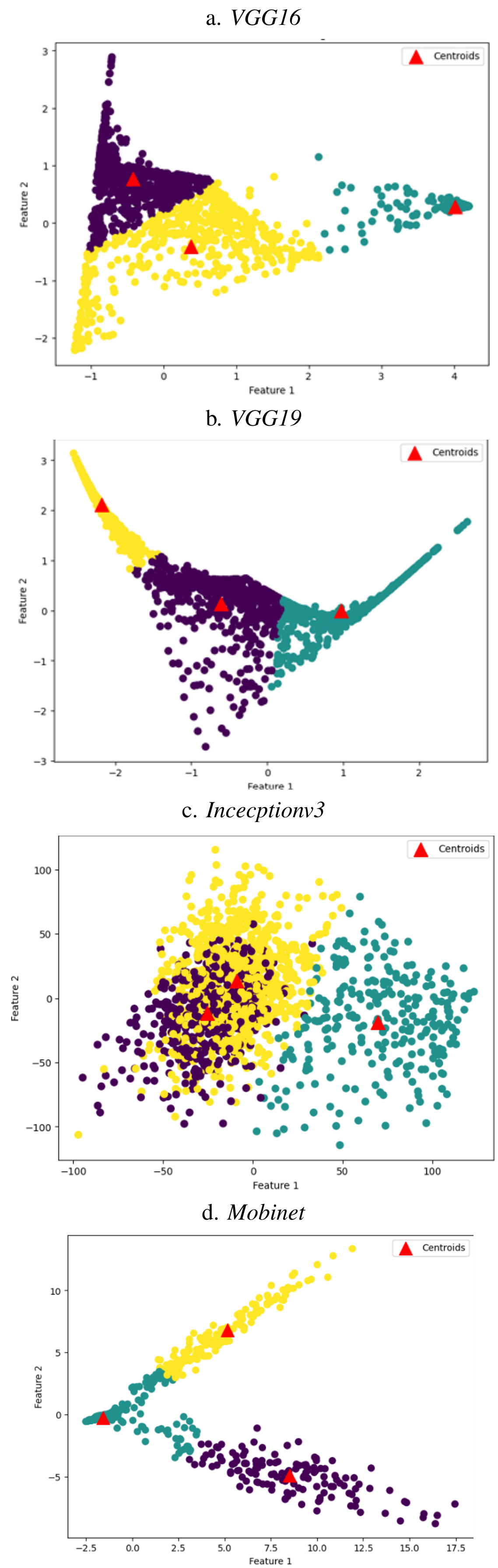


Fig. 2. Feature cluster of PC1 and PC2 in feature space

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TABLE III
ACCURACY AFTER FINE TUNING

| Model | Accuracy |
|---|---|
| VGG16 | 61.29% |
| VGG19 | 58.84% |
| IncecptionV3 | 58.71% |
| MobiNet | 67.82% |

fine-tuning.The network confusion matrices show ambiguity with knife and handheld images as presented in Fig.5.

a.*vgg16*

| | precision | recall | f1-score | support |
|---|---|---|---|---|
| knife | 0.21 | 0.30 | 0.24 | 311 |
| guns | 0.34 | 0.11 | 0.17 | 450 |
| handheld | 0.50 | 0.61 | 0.55 | 750 |
| accuracy | | | 0.40 | 1511 |
| macro avg | 0.35 | 0.34 | 0.32 | 1511 |
| weighted avg | 0.39 | 0.40 | 0.37 | 1511 |

b. *vgg19*

| | precision | recall | f1-score | support |
|---|---|---|---|---|
| knife | 0.19 | 0.27 | 0.22 | 311 |
| guns | 0.30 | 0.21 | 0.25 | 450 |
| handheld | 0.50 | 0.50 | 0.50 | 750 |
| accuracy | | | 0.36 | 1511 |
| macro avg | 0.33 | 0.32 | 0.32 | 1511 |
| weighted avg | 0.38 | 0.36 | 0.37 | 1511 |

c. *incecptionv3*

| | precision | recall | f1-score | support |
|---|---|---|---|---|
| knife | 0.20 | 0.47 | 0.28 | 311 |
| guns | 0.32 | 0.09 | 0.14 | 450 |
| handheld | 0.51 | 0.45 | 0.48 | 750 |
| accuracy | | | 0.35 | 1511 |
| macro avg | 0.34 | 0.34 | 0.30 | 1511 |
| weighted avg | 0.39 | 0.35 | 0.34 | 1511 |

d. *mobinet*

| | precision | recall | f1-score | support |
|---|---|---|---|---|
| knife | 0.21 | 0.32 | 0.25 | 311 |
| guns | 0.26 | 0.21 | 0.23 | 450 |
| handheld | 0.49 | 0.44 | 0.47 | 750 |
| accuracy | | | 0.35 | 1511 |
| macro avg | 0.32 | 0.32 | 0.32 | 1511 |
| weighted avg | 0.37 | 0.35 | 0.35 | 1511 |

Fig. 3. Performance matrices for fine tuned models

This is because the dataset chosen in our work had overlapping images between knife and handheld class.Due to this network was ambiguous between the two which is clearly seen with low F1 score with hand held weapons as presented in Fig.3

After fine tuned models and their performance matrices were evaluated. The results were not promising, All the selected model performed poor than expected with chosen dataset in our experiment. For example Looking at Fig.3 models have performed not as desired and clearly struggling to generalize understanding of a knife because network was not able to generalize knife held in hand and knife as subject in image itself.Now there was a need for improvement in accuracy of these networks. To do so one way could be as presented by research mentioned our literature review is to in general add mode neural layers and more convolution layers. But this also increase the computational cost and memory cost by a massive factor. Looking at the VGG16 with 138 Million parameters and VGG19 with 144 Million parameters there is an increase in complexity by a factor of 1.06 yet yielding no significant performance improvement over each other. Looking the same way at incecptionV3 have around 24 Million parameters and Mobinet we used has only 5.3 Million parameters. They are significantly smaller that VGG design with reduced complexity factor by massive 26.7.Hence,in conclusion just by adding more layers and increasing complexity factor won't solve the issue and even if it did, not without increasing significant computational cost. With surveillance as goal, It become obvious to take care of edge device running these algorithm on site for real-time detection with least possible power consumption. Therefore it was crucial to experiment with machine learning algorithms in cooperation with deep learning networks to see if it enhance performance as hybrid design with minimal increase in complexity factor and power consumption. Following fine-tuning, the deep learning models were integrated with k-means clustering algorithm. Kmeans [11] algorithm was selected for three broad reasons. Firstly, because clustering is an unsupervised algorithm and allowed us to train over unlabeled feature extracted by fine tuned models, Secondly it can be trained multiple times without retraining fine tuned models along with it. This becomes huge advantage considering edge device performing surveillance in real-time with our approach, providing onsite learning with minimal computational cost, and finally suppose during production, if the fine-tuned model encounter an unfamiliar entity in image , the extracted feature descriptor will significantly deviate from the learned cluster in feature space like an anomaly. Such outlier will be alarming indicator for network's limitation in knowledge and incompleteness in application of real world, many times that cannot be replicated in training dataset. With limitation known, it can be soon rectified with supervised human intervention in production. Each fine tuned network yield 16-dimensional feature except inception V3 yielding 512-dimensional feature. It was not necessary that all the dimensions are necessary to represent feature space,Principal Component Analysis (PCA) [12] was used to purge dimensions that have minuscule impact on feature, meaning feature descriptors were nearly invariant to change occurring in those dimensions and hence first two principle components were selected to perform clustering. Clustering was employed to identify inherent patterns and groupings, facilitating the creation of distinct clusters representing different classes of weapon imagery.

We observed models to perform better as presented. Mobinet acheived highest performance as presented in TableIV.Using K-means clustering along with fine tuned models as feature extractor,the hybrid design was able to generalize the hypothesis further. Incecption have performed worse with

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TABLE IV
ACCURACY AFTER USING K-MEANS CLUSTERING OVER FEATURE

| Model | Accuracy |
|---|---|
| VGG16 | 66.86% |
| VGG19 | 73.10% |
| IncecptionV3 | 53% |
| MobiNet | 98.34% |

hybrid design. Without hybrid design it failed to generalize the hypothesis to achieve reasonable accuracy.

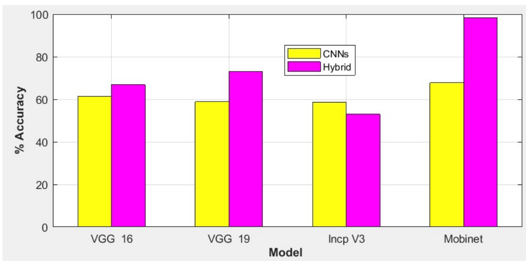


Fig. 4. Performance comparision plot

Hypothesis of a network is its neural weights[13] generalization makes a learning algorithm able to search solutions with least error even if there are similar features of two different entity.It can classify more easily even if two objects are very similar.This gives learning algorithm a robust design. In Fig.2 we present cluster formation of first two principle components of feature vector extracted by fine tuned networks. Using the proposed hybrid architecture to identify weapons on CCTV footage, we can infer from these feature space plots that our model is far better than Inception V3 network. Excellent work is being done in projecting various weapons in feature space using mobile net and VGG versions. So, it is not appropriate to use incecptionV3 for this purpose.

There are three major classifications of weapons in the OD-weapon detection datasets that we utilised.A type of hand carried weapon was one of them. Knives were also shown on the hand-held weaponry. This led the network to identify knives as belonging to a class that includes both knives and hand-held weapons. Impact of this dichotomy can be seen in the confusion matrices presented in Fig.5

Our results highlight how well deep learning and machine learning approaches work together to detect weapon pictures,We conclude that MobiNet hybrid design outperform all the other network in the task of weapon detection in CCTV footage as shown in Fig.4.This is an ideal decision because the network was built for low-powered edge devices. A camera with MobiNet installed may detect threats in real time based on what's seen, exhibiting remarkably high accuracy.

## VII. CONCLUSION AND FUTURE REMARK

We find that MobiNet hybrid architecture outperforms all other networks in performing the task of weapon recognition in a given image. Our results demonstrate the effective combination of deep learning and machine learning techniques

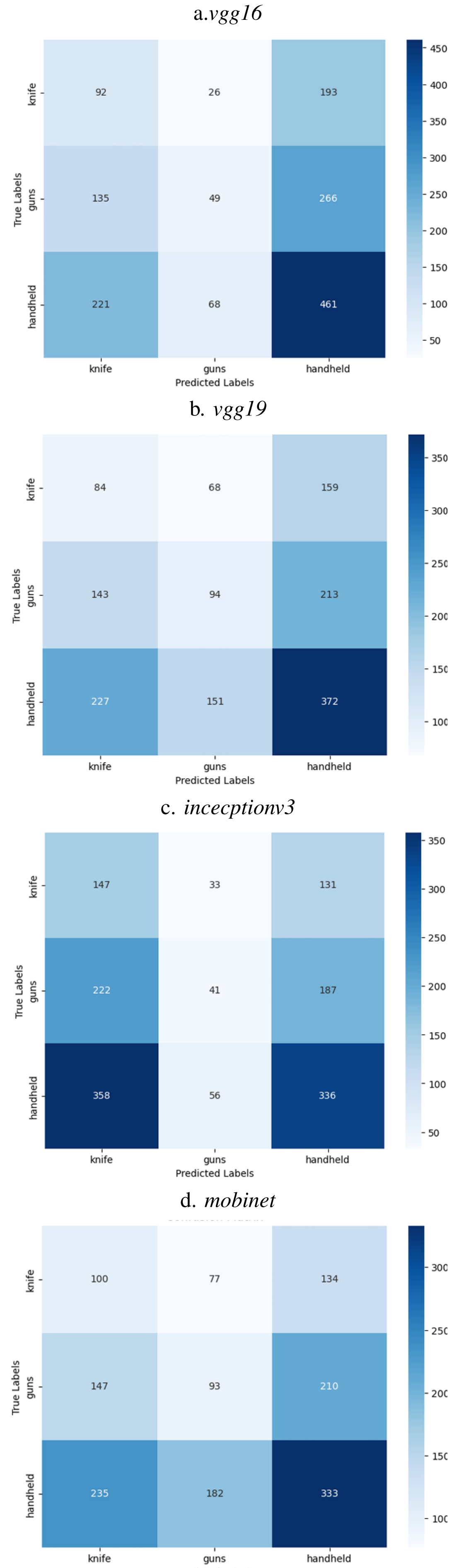


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Fig. 5. Confusion Matrices(without kmeans)

in detecting photographs of weapons. This is an excellent option because the network was created with low-powered edge devices in mind. A camera that can identify threats in real time may be equipped with MobiNet, which operates with exceptional precision. We determine that MobiNet, out of the three designs considered for the weapon detection work in Inception V3, VGG16/19, is the best design based on the performance matrices. To improve MobiNet's capacity to distinguish between weapons and non-weapon objects carried by individuals, more research will be needed in the future. Based on the performance matrices, we determine that MobiNet is the optimal design among the Inception V3 and VGG16/19 models that were assessed for the weapon identification task. However, further research is needed to improve MobiNet's capacity to distinguish between weapons and non-weapon objects held in a person's hand in the provided film, and future work on potential solutions to this problem is also necessary.

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